BETOBULLET

14 HP · EURORACK FORMAT
FOR VCV RACK 2



ZEMUN

DUAL PEAK MULTIMODE FILTER

USER MANUAL · v2.0.0 · October 2026

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Zemun is a dual peak multimode state variable filter: two resonant cores, ten responses from double low-pass to double high-pass, a span control that pulls the peaks up to eight octaves apart, a balance between them and audio-rate self and cross modulation.

1. INTRODUCTION

Two state variable filter cores sit at the heart of Zemun. Depending on the mode they run in series or in parallel, and their low-pass, band-pass, high-pass and notch outputs are combined into ten different responses. Both cores follow the FREQ knob and the V/OCT input; SPAN moves them apart symmetrically, BALANCE shifts the resonance from one peak to the other, and TITO lets the filter modulate itself at audio rate.

The cores are nonlinear: their states saturate softly, so high resonance and hot inputs distort musically instead of clipping, and above about 8 on the RESO knob the filter self-oscillates as a tuned sine source. Processing runs 2× oversampled.

Zemun recreates the published behaviour of the Xaoc Devices Belgrad; the DSP is an independent design.

INSTALLATION

Copy `Zemun-2.0.0-win-x64.vcvplugin` to `Documents/Rack2/plugins` and restart VCV Rack. The module is listed under the brand **betobullet**.

2. PANEL OVERVIEW



1. **FREQ**
   Centre frequency of both peaks, 4 Hz to 28 kHz.
2. **LEVEL**
   Input gain. Above about 40 % the input stage saturates softly.
3. **SPAN**
   Distance between the two peaks, 0 to 8 octaves, on a non-linear scale.
4. **RESO**
   Resonance. Self-oscillation starts around 8.
5. **BALANCE**
   Moves the resonance between the lower and the higher peak.
6. **MODE**
   Ten responses: LL LB NL NN LH BB HL HN BH HH.
7. **TITO**
   Up: self modulation (SM). Middle: off. Down: cross modulation (XM).
8. **FM amount**
   Slider: depth of the FM input. It lights with the CV it passes.
9. **SPAN amount**
   Slider: depth of the SPAN CV input. It lights with the CV.
10. **RESO CV**
    0 to 8 V sweeps the whole resonance range.
11. **V/OCT**
    1 V/octave pitch input; both peaks track.
12. **BALANCE CV**
    −5 V to +5 V sweeps the whole balance range.
13. **FM**
    Exponential FM, scaled by slider (8).
14. **INPUT**
    Audio input, ±5 V nominal.
15. **OUTPUT**
    Filter output.
16. **SPAN CV**
    ±5 V covers the whole 8 octaves with slider (9) fully up.

3. FILTER MODES

The MODE switch **(6)** selects how the two cores are combined. The first letter refers to the lower peak, the second to the higher one.

MODE	STRUCTURE	RESPONSE
LL	LP → LP (series)	Double-slope low-pass. With SPAN at 0 a steep 4-pole low-pass; opening SPAN gives a low-pass with two resonant humps.
LB	LP + BP (parallel)	Low-pass with an extra band-pass peak above it.
NL	notch → LP	Low-pass with a notch inside the pass band.
NN	notch → notch	Double notch: two rejected bands, phaser-like with modulation.
LH	LP + HP (parallel)	Band reject between the peaks.
BB	BP + BP (parallel)	Double band-pass: two formant-like peaks, vowel sounds.
HL	HP → LP	Band-pass with a flat top between wide peaks.
HN	HP → notch	High-pass with a notch.
BH	BP + HP (parallel)	High-pass with an extra band-pass peak below it.
HH	HP → HP (series)	Double-slope high-pass, the mirror of LL.

4. CONTROLS IN DETAIL

FREQ AND V/OCT

FREQ **(1)** sets the centre between the two peaks, from 4 Hz to 28 kHz. The V/OCT input **(11)** adds to it at 1 V/octave, and FM **(13)** adds an exponential modulation scaled by its slider **(8)**.

SPAN

SPAN **(3)** moves the two peaks apart symmetrically around the centre, from 0 (both peaks on top of each other) to 8 octaves. The scale is deliberately non-linear: fine near zero, wide at the end. The SPAN CV **(16)**, scaled by its slider **(9)**, adds up to ±8 octaves.

RESO AND BALANCE

RESO **(4)** sets the resonance of both cores. Around 8 the filter starts to self-oscillate, tuned to the cutoff and tracking V/OCT; higher settings add distortion. BALANCE **(5)** takes resonance away from one core and gives it to the other: fully left emphasises the lower peak, fully right the higher one.

LEVEL

LEVEL **(2)** is the input gain. Low settings keep the filter clean; higher settings drive the input stage and the cores into soft saturation.

TITO

TITO **(7)** adds audio-rate modulation of the cores' frequencies by their own band-pass signals. In **SM** each core modulates itself; in **XM** each core modulates the other. With resonance this ranges from growl to chaos.

POLYPHONY

Zemun is polyphonic: it runs one filter per channel of INPUT or V/OCT, up to 16.

5. PATCH IDEAS

- **Sine oscillator**: no input, RESO above 8, a pitch into V/OCT.
- **Talking filter**: mode BB, SPAN around noon, RESO 6, a slow random CV into SPAN CV.
- **Thick bass**: mode LL, SPAN low, LEVEL high for saturation, an envelope into FM.
- **Phaser**: mode NN, a triangle LFO into FM, RESO low.
- **Controlled chaos**: TITO in XM, RESO near 8, sweep BALANCE.

6. SPECIFICATIONS

ITEM	VALUE
Width	14 HP
Cores	2 nonlinear state variable filters, 2× oversampled
Modes	10
Frequency	4 Hz – 28 kHz, 1 V/oct
Span	0 – 8 octaves
Self-oscillation	from RESO ≈ 8, tuned
Inputs	INPUT, V/OCT, FM, SPAN, RESO, BALANCE
Output	±12 V max, soft saturation
Polyphony	up to 16 channels

CREDITS

Design, DSP and panel: betobullet. Behaviour modelled on the Xaoc Devices Belgrad operator's manual; Belgrad is a product of Xaoc Devices and this is an independent recreation. Fonts: Black Ops One, Space Mono, Space Grotesk (SIL OFL).